Assignment:

1. Write a Python program to sum all the items in a list.

```
def sum(mylist):
  total = 0
  for s in mylist:
    total += s
  return total
mylist = [1,2,3,4]
print(f"Sum of {mylist} : {sum(mylist)}")
```

```
Sum of [1, 2, 3, 4] : 10
```

2. Write a Python program to get the largest and smallest number from a list without builtin functions.

```
mylist = [1,2,3,4,5]
min = mylist[0]
max = mylist[0]
for i in range(1,len(mylist)):
  if min > mylist[i]:
    min = mylist[i]
  if max < mylist[i]:
    max = mylist[i]
print(f"Maximum from {mylist} : {max}")
print(f"Minimum from {mylist} : {min}")
```

```
Maximum from [1, 2, 3, 4, 5] : 5
Minimum from [1, 2, 3, 4, 5] : 1
```

3. Write a Python program to find duplicate values from a list and display those.

```
mylist = [1,2,2,3,3,4,44,4,5]
list2 = set()
list3 = set()
for s in mylist:
  if s in list2:
    list3.add(s)
  else:
    list2.add(s)
print(mylist)
print("Duplicates : ", list3)
```

```
[1, 2, 2, 3, 3, 4, 44, 4, 5]
Duplicates :  {2, 3, 4}
```

4. Write a Python program to split a given list into two parts where the length of the first part of the list is given. Original list: [1, 1, 2, 3, 4, 4, 5, 1] Length of the first part of the list: 3 Splitted the said list into two parts: ([1, 1, 2], [3, 4, 4, 5, 1])

```
def split_list(og, len):
  return og[:len], og[len:]

og = [1, 1, 2, 3, 4, 4, 5, 1]
len = 3
split_list(og, len)
```

```
([1, 1, 2], [3, 4, 4, 5, 1])
```

5. Write a Python program to traverse a given list in reverse order, and print the elements with the original index. Original list: ['red', 'green', 'white', 'black'] Traverse the said list in reverse order: black white green red

```
og = ['red', 'green', 'white', 'black']
new = []
for s in og[::-1]:
  new.append(s)
print(new)
```

```
['black', 'white', 'green', 'red']
```

Assignment:

1. Write a Python program and calculate the mean of the below dictionary. test_dict = {"A" : 6, "B" : 9, "C" : 5, "D" : 7, "E" : 4} Output: 6.2

```
test_dict = {"A": 6, "B": 9, "C": 5, "D": 7, "E": 4}

mean_value = sum(test_dict.values()) / len(test_dict)

print(mean_value)
```

2.Write a Python script to concatenate the following dictionaries to create a new one. Sample Dictionary : dic1={1:10, 2:20} dic2={3:30, 4:40} dic3={5:50,6:60} Expected Result : {1: 10, 2: 20, 3: 30, 4: 40, 5: 50, 6: 60}

```
dic1={1:10, 2:20}
dic2={3:30, 4:40}
dic3={5:50,6:60}
mydict = dic1.copy()
mydict.update(dic2)
```

```
mydict.update(dic3)
print(mydict)
```

```
{1: 10, 2: 20, 3: 30, 4: 40, 5: 50, 6: 60}
```

3.Write a Python program to get the key, value and item in a dictionary. input:dict_num = {1: 10, 2: 20, 3: 30, 4: 40, 5: 50, 6: 60}

```
num = {1: 10, 2: 20, 3: 30, 4: 40, 5: 50, 6: 60}
print("Key", "Value")
for key, values in num.items():
  print(key, values)
```

```
Key Value
1 10
2 20
3 30
4 40
5 50
6 60
```

4.Write a Python program to get the key, value and item in a dictionary. Input: input_dict = {1: 10, 2: 20, 3:None, 4: 40, 5: None, 6: 60}

```
input_dict = {1: 10, 2: 20, 3:None, 4: 40, 5: None, 6: 60}
dict = {key: value for key, value in input_dict.items() if value is not None}
print(f"Dictionary with Empty Items Dropped: {dict}")
```

```
Dictionary with Empty Items Dropped: {1: 10, 2: 20, 4: 40, 6: 60}
```

Assignment:

1. Write a Python program to find the number of times 4 appears in the tuple. Input: tuplex = (2, 4, 5, 6, 2, 3, 4, 4, 7) Output: 3

```
tuplex = (2, 4, 5, 6, 2, 3, 4, 4, 7 )
print(tuplex.count(4))
```

```
3
```

2.Write a Python program to convert a list to a tuple. Input: listx = [5, 10, 7, 4, 15, 3] Output: (5, 10, 7, 4, 15, 3)

```
listx = [5, 10, 7, 4, 15, 3]
mytuple = tuple(listx)
print(mytuple)
```

```
(5, 10, 7, 4, 15, 3)
```

3. Write a Python program to calculate the sum of the numbers in a given tuple. Input: tuples_list = [(1, 2), (3, 4), (5, 6)] Output:

```
tuples_list = ((1, 2), (3, 4), (5, 6))
sum = 0
for s in tuples_list:
  for i in s:
    sum += i
print(sum)
```

```
21
```

4.Write a python program and iterate the given tuples Input: employee1 = ("John Doe", 101, "Human Resources", 60000) employee2 = ("Alice Smith", 102, "Marketing", 55000) employee3 = ("Bob Johnson", 103, "Engineering", 75000)

```
employee1 = ("John Doe", 101, "Human Resources", 60000)
employee2 = ("Alice Smith", 102, "Marketing", 55000)
employee3 = ("Bob Johnson", 103, "Engineering", 75000)
employees = [employee1, employee2, employee3]
for e in employees:
  name, employee_id, department, salary = e
  print("Employee Records:")
  print("Name: ", name)
  print("Employee ID: ", employee_id)
  print("Department: ", department)
  print("Salary: ", salary)
  print()
```

```
Employee Records:
Name:  John Doe
Employee ID:  101
Department:  Human Resources
Salary:  60000

Employee Records:
Name:  Alice Smith
Employee ID:  102
Department:  Marketing
Salary:  55000

Employee Records:
Name:  Bob Johnson
Employee ID:  103
Department:  Engineering
Salary:  75000
```

Assignment:

1. Write a Python program to Get Only unique items from two sets. Input: set1 = {10, 20, 30, 40, 50} set2 = {30, 40, 50, 60, 70} Output: {70, 40, 10, 50, 20, 60, 30}

```
set1 = {10, 20, 30, 40, 50}
set2 = {30, 40, 50, 60, 70}
unique = set1.union(set2)
print(unique)
```

```
{70, 40, 10, 50, 20, 60, 30}
```

2. Write a Python program to Return a set of elements present in Set A or B, but not both. Input: set1 = {10, 20, 30, 40, 50} set2 = {30, 40, 50, 60, 70} Output: {20, 70, 10, 60}

```
set1 = {10, 20, 30, 40, 50}
set2 = {30, 40, 50, 60, 70}
res = set1 ^ set2
print(res)
```

```
{20, 70, 10, 60}
```

3. Write a Python program to Check if two sets have any elements in common. If yes, display the common elements. Input:

set1 = {10, 20, 30, 40, 50} set2 = {60, 70, 80, 90, 10}

Output: {10}

```
set1 = {10, 20, 30, 40, 50}
set2 = {60, 70, 80, 90, 10}
res = set1.intersection(set2)
print(res)
```

```
{10}
```

4. Write a Python program to Remove items from set1 that are not common to both set1 and set2. Input: set1 = {10, 20, 30, 40, 50} set2 = {30, 40, 50, 60, 70} Output: {40, 50, 30}

```
set1 = {10, 20, 30, 40, 50}
set2 = {30, 40, 50, 60, 70}
set1.intersection_update(set2)
print(set1)
```

```
{40, 50, 30}
```